SECTION 02:
DEEP LEARNING (DL)

Advancing Beyond Traditional Machine Learning

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DEFINITION OF DEEP LEARNING

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Deep Learning (DL) is a subset of Machine Learning (ML) that uses Artificial Neural Networks (ANNs) with multiple hidden layers to automatically learn complex patterns from large amounts of data.

Unlike traditional Machine Learning, Deep Learning does not require manual feature extraction. It automatically learns features from the input data and improves its performance through training.

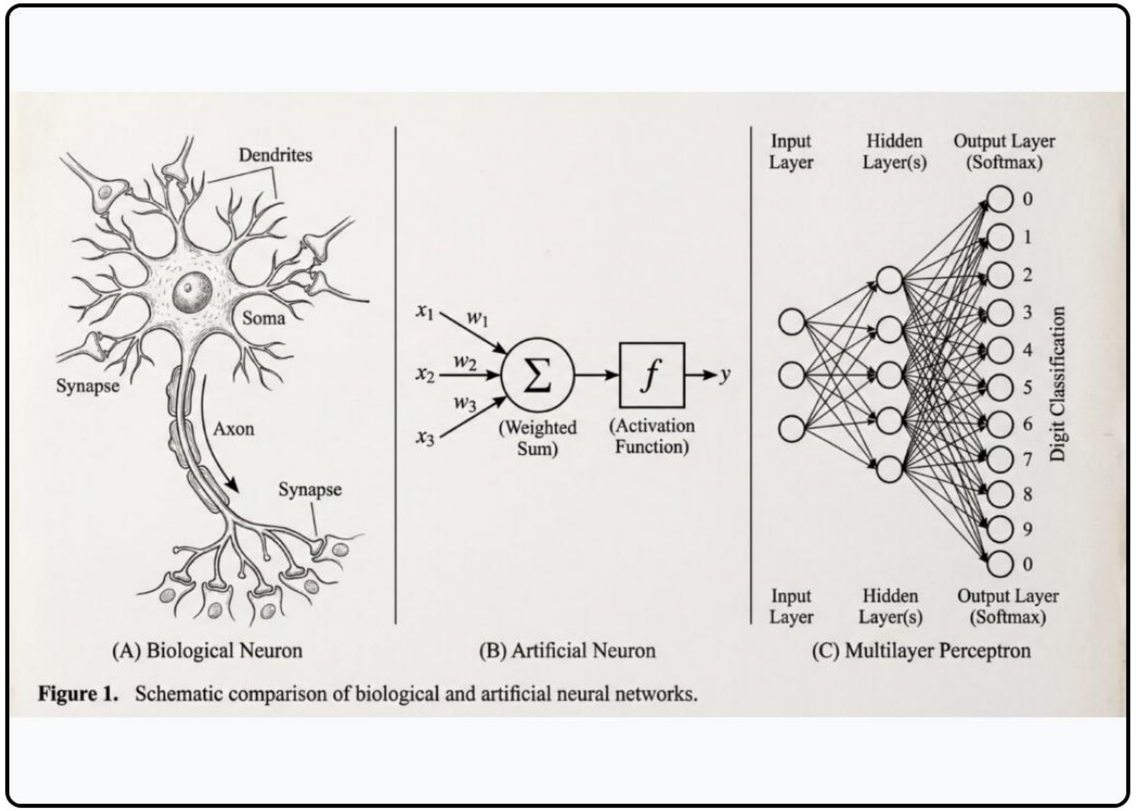
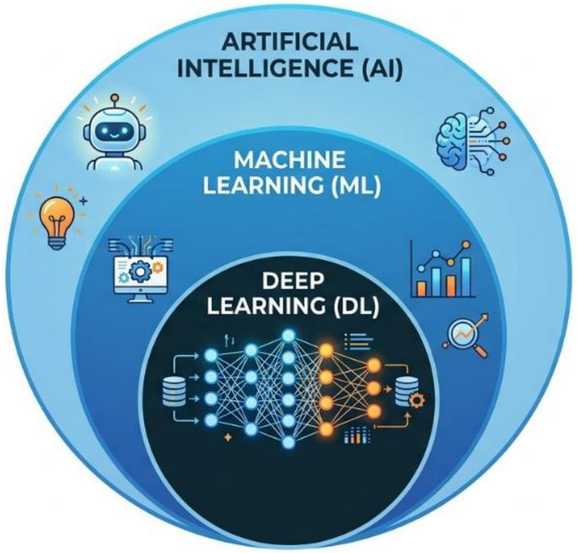


Figure 1. Schematic comparison of biological and artificial neural networks.

RELATIONSHIP: AI, ML, AND DL



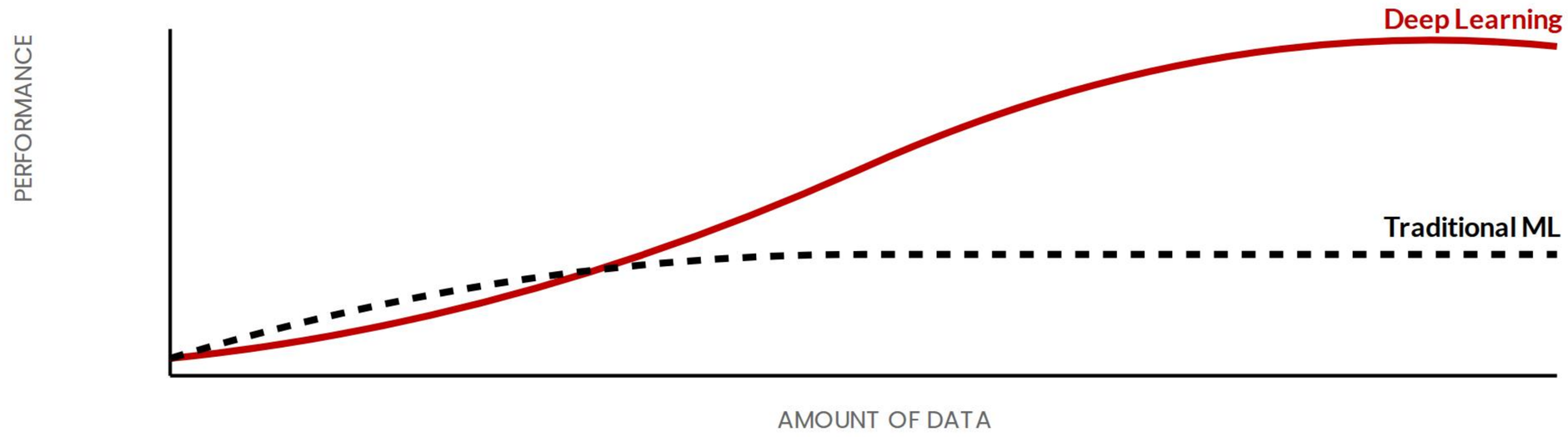
HIERARCHY EXPLANATION

- AI is the broad field of making machines intelligent.
- ML is a subset of AI where machines learn from data.
- DL is a subset of ML that uses deep neural networks with many hidden layers.

Note: Deep Learning represents the most specialized and automated form of AI currently in use.

WHY DEEP LEARNING?

Traditional Machine Learning has limitations when dealing with **large and complex datasets**. Deep Learning overcomes these limitations by **automatically learning** important features from data.



Deep Learning performance continues to scale with more data, while traditional ML plateaus.

FEATURES AND ADVANTAGES

KEY FEATURES

- Learns automatically from data.
- Uses ANN architecture.
- Multiple hidden layers.
- Automatic feature extraction.

PERFORMANCE

- Handles large datasets efficiently.
- High accuracy for complex tasks.
- Supports real-time applications.

DATA UTILITY

- Processes structured and unstructured data.
- No manual engineering required.
- Learns complex patterns.

DISADVANTAGES AND CONSTRAINTS

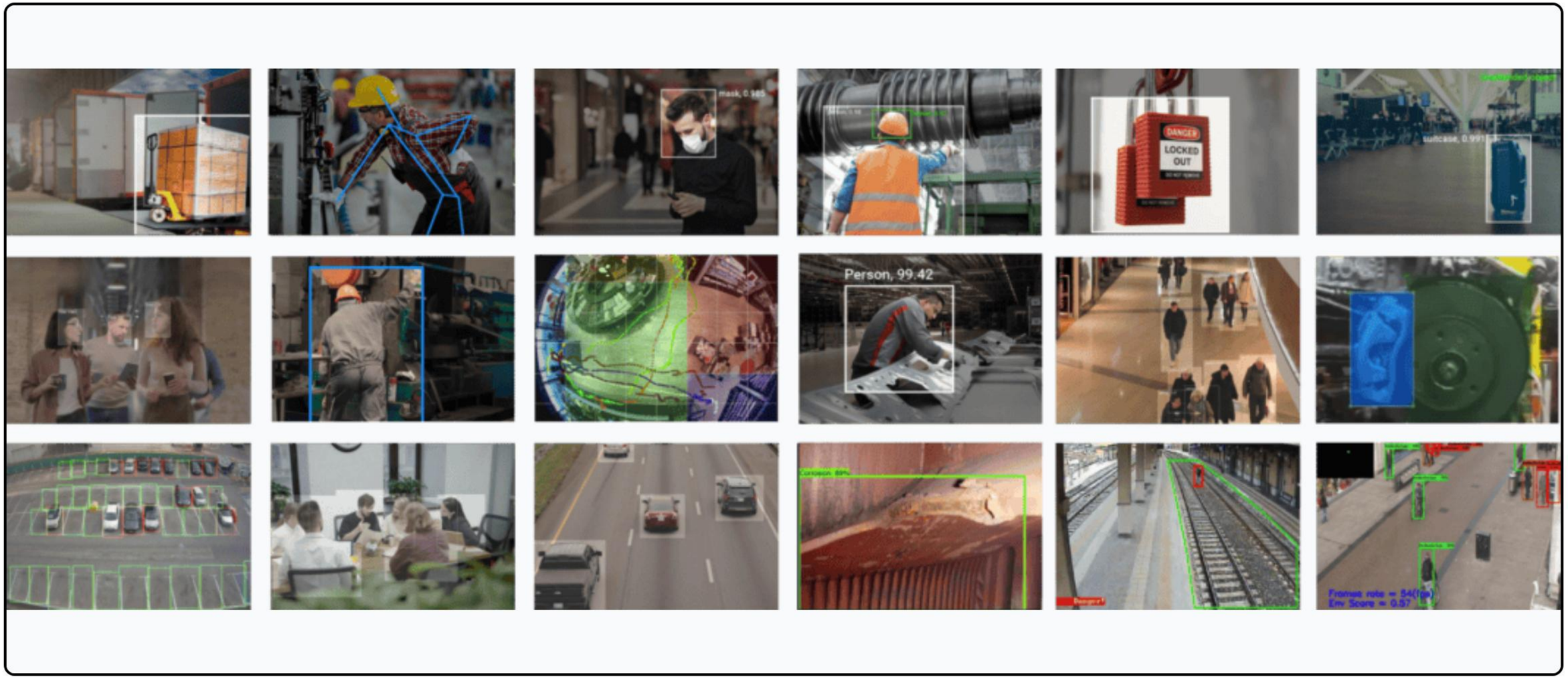
- **Data Intensive:** Requires a large amount of training data.
- **Computation:** Needs high computational power (GPU/TPU).
- **Time:** Training takes a long time.
- **Black Box:** Difficult to interpret (Black Box Model).
- **Cost:** High implementation cost.



APPLICATIONS OF DEEP LEARNING

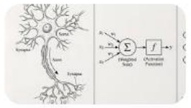
- Image & Face Recognition
- Speech Recognition
- NLP & Machine Translation
- Chatbots & Virtual Assistants
- Medical Diagnosis
- Autonomous Vehicles
- Fraud Detection
- Recommendation Systems

INDUSTRY USE CASES



Deep Learning is widely used in high-stakes fields like healthcare, security, and transportation.

IMAGE SOURCES



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